

Project 4: The Setup

Model: 1D shallow ReLU network, slopes fixed at $w_j = 1$:

$$f(x) = \sum_{j=1}^m a_j \sigma(x - b_j), \quad \sigma(z) = \max(0, z), \quad x \in [-1, 1]$$

This is a **piecewise linear** function. Between bias points, the function is linear with a fixed slope. At each b_j , the slope changes by a_j .

Training: Gradient flow on MSE loss over the continuous domain:

$$\mathcal{L} = \frac{1}{2} \int_{-1}^1 (f(x) - f^*(x))^2 dx$$

where $f^*(x)$ is a smooth target (e.g., sum of sinusoids).

The surprise: Even with $m = 100$ neurons, the biases b_j **collapse** to only ~ 6 – 8 distinct values. The network discovers it only needs a few kinks.

Project 4: The Gradient Flow ODEs — Amplitudes

Amplitude dynamics:

$$\dot{a}_j = -\frac{\partial \mathcal{L}}{\partial a_j} = -\int_{-1}^1 (f(x) - f^*(x)) \sigma(x - b_j) dx$$

Interpretation: $\dot{a}_j$ is the **negative correlation** between the residual error $f - f^*$ and the ReLU basis function $\sigma(x - b_j)$.

- If the residual is positive where $\sigma(x - b_j)$ is large (to the right of b_j), then a_j decreases.
- If the residual is negative there, a_j increases.
- The amplitude adjusts to reduce the projected error along that ReLU feature.

Project 4: The Gradient Flow ODEs — Biases

Bias dynamics:

$$\dot{b}_j = -\frac{\partial \mathcal{L}}{\partial b_j} = a_j \int_{-1}^1 (f(x) - f^*(x)) \sigma'(x - b_j) dx = a_j \int_{b_j}^1 (f(x) - f^*(x)) dx$$

Interpretation: $\dot{b}_j$ is proportional to the **integrated residual** to the right of b_j , weighted by the neuron's amplitude a_j .

Key insight: If two biases b_i and b_j are close, they see nearly the same integrated residual:

$$\int_{b_i}^1 (f - f^*) dx \approx \int_{b_j}^1 (f - f^*) dx$$

Their dynamics **synchronize** — they move together.

Project 4: The Empirical Phenomenon

Observation (supported by numerical experiments, formal proof is Open Problem 4.1):

Let $f^*(x)$ be a target on $[-1, 1]$. Let k be the number of kinks in the optimal m -term piecewise linear L^2 -approximation of f^* . If $m > k$, then under gradient flow:

1. The biases $b_j(t)$ appear to converge to at most k distinct limit points.
2. The amplitudes of biases sharing a limit point sum to the combined weight.
3. The network output converges to the optimal k -kink approximation of f^* .

Status: This is a **conjecture** supported by numerical evidence. Proving it — the cluster count, convergence, and error bounds — is the goal of Open Problems 4.1–4.3.

Chapter 11. Formal proof is open — this is the research contribution expected from the project.

Project 4: Why Collapse Happens

Fundamental reason: A ReLU network with sorted biases $b_1 < \dots < b_m$ is a piecewise linear function with kinks at the b_j . If the target f^* can be well-approximated with k kinks, then $m - k$ neurons are redundant.

The redundancy is explicit: Two biases at the same location $b_i = b_j$ with amplitudes a_i, a_j produce **exactly** the same function as one bias at b_i with amplitude $a_i + a_j$:

$$a_i \sigma(x - b_i) + a_j \sigma(x - b_i) = (a_i + a_j) \sigma(x - b_i)$$

Gradient flow exploits this: The loss landscape has **flat directions** corresponding to moving redundant biases. Along these directions, the loss doesn't change (to first order). Second-order effects drive biases toward the active ones.

Project 4: Synchronization — Mathematical Analysis

Suppose b_i and b_j are close: $|b_i - b_j| = \varepsilon \ll 1$.

Difference in bias velocities:

$$\dot{b}_i - \dot{b}_j = a_i \int_{b_i}^{b_j} (f - f^*) dx + (a_i - a_j) \int_{b_j}^1 (f - f^*) dx$$

- **First term:** $O(\varepsilon)$ — the integral over a tiny interval. Vanishes as biases approach.
- **Second term:** Proportional to $(a_i - a_j)$. The amplitude difference drives further synchronization.

Positive feedback: Close biases $\rightarrow$ similar dynamics $\rightarrow$ stay close $\rightarrow$ amplitudes equalize $\rightarrow$ biases stay synchronized. This feedback loop **locks** nearby biases together.

Project 4: Merging & Stability

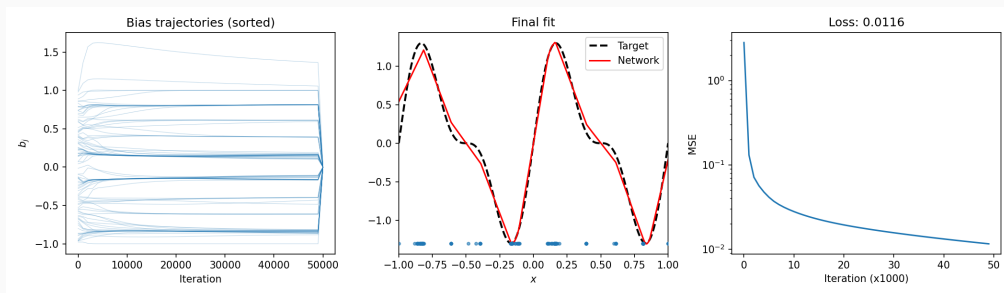
Once merged ($b_i = b_j$): The two neurons are **exactly equivalent** to one neuron.

Stability of the merged state:

- **First order:** The gradient with respect to separating the biases is zero — no incentive to split.
- **Second order:** The Hessian along the separation direction is **positive semidefinite** — splitting increases or maintains the loss.
- **Conclusion:** The merged state is **neutrally stable**. Once biases merge, gradient flow won't separate them.

Overall process: Biases start spread out. Nearby pairs synchronize and merge. The merged groups themselves interact and may merge further. The process continues until only k distinct values remain, where k is determined by the target complexity.

Project 4: Empirical Demonstration



$m = 100$ neurons, target $f^*(x) = \sin(2\pi x) + 0.5 \sin(4\pi x)$ on $[-1, 1]$.

- **Left:** Sorted bias trajectories over 50k iterations. Initial uniform spread $\rightarrow$ tight clusters.
- **Center:** Final fit (red) vs. target (dashed). Dots mark bias locations.
- **Right:** MSE loss on log scale. Smooth decay — no discrete jumps.

Full code in Chapter 11, Section 4.3.

Project 4: What We Learn from the Experiment

Key observations:

1. **Continuous clustering:** Biases approach each other smoothly — there are no discrete collision events. The process is a smooth dynamical flow.
2. **~6–8 clusters:** For a target with 8 curvature changes ($(f^*)''$ sign flips), biases organize into approximately that many groups. This suggests a connection between target complexity and effective architecture size.
3. **Non-uniform cluster sizes:** Some locations attract many biases (high “demand” for a kink), others few. The distribution of cluster sizes may reflect local approximation difficulty.
4. **Smooth loss decay:** Despite massive structural reorganization (100 biases $\rightarrow$ ~8 effective), the loss decreases smoothly.

Project 4: Open Problem 4.1 — Cluster Count

The question: For a given f^* and width m , exactly how many bias clusters emerge?

Conjecture:

$$\lim_{m \rightarrow \infty} C(m, f^*) = \#\{x \in [-1, 1] : (f^*)''(x) = 0 \text{ and changes sign}\}$$

The number of clusters is bounded by the number of **curvature changes** (inflection points) of the target.

Why curvature changes? Each inflection point is where the optimal piecewise linear approximation **must** place a kink. The conjecture says this lower bound is asymptotically tight.

Specific goals:

1. Prove $C(m, f^*) \leq C_{\max}(f^*)$ independent of m .
2. Is convergence to clusters **finite-time** or **asymptotic**?
3. Does the count depend on curvature **amplitude** or only sign pattern?

Project 4: Open Problem 4.2 — Higher Dimensions

The question: Does parameter collapse generalize to $\mathbb{R}^d$ for $d \geq 2$?

General model: $f(x) = \sum_{j=1}^m a_j \sigma(w_j^\top x + b_j), x \in \mathbb{R}^d$.

In 1D, a neuron defines a **point** b_j (the kink location). In $\mathbb{R}^d$, a neuron defines a **hyperplane** $\{x : w_j^\top x + b_j = 0\}$.

Collapse could mean:

1. **Hyperplane collapse:** $(w_i, b_i) \propto (w_j, b_j)$ — same hyperplane.
2. **Directional collapse:** $w_i \propto w_j$ but biases differ — parallel hyperplanes.
3. **Mixed:** Only biases merge, directions remain distinct, or vice versa.

Approach: Start with $d = 2$ and simple targets. Visualize hyperplane configurations.

Project 4: Open Problem 4.3 — Provable Pruning

The question: Can collapsed neurons be **provably** removed without hurting performance?

The pruning procedure: After training, biases cluster into k groups. For each cluster $\mathcal{C}_\ell$ with centroid $\bar{b}_\ell$:

1. Replace all biases in $\mathcal{C}_\ell$ with $\bar{b}_\ell$.
2. Sum their amplitudes: $\bar{a}_\ell = \sum_{j \in \mathcal{C}_\ell} a_j$.
3. Pruned network: $\tilde{f}(x) = \sum_{\ell=1}^k \bar{a}_\ell \sigma(x - \bar{b}_\ell)$.

The guarantee to prove: If intra-cluster distance $\leq \varepsilon$, then:

$$\|\tilde{f} - f\|_\infty \leq \varepsilon \sum_{j=1}^m |a_j|$$

The key challenge: **bound** $\sum |a_j|$ uniformly during training.

Project 4: Broader Connections — Lottery Ticket

Lottery Ticket Hypothesis (Frankle & Carbin, 2019): Within a randomly initialized network, there exists a sparse subnetwork that trains to match the full network's performance.

Connection to bias collapse:

- LTH identifies sparsity **retrospectively**: train, prune, rewind, retrain.
- Bias collapse creates sparsity **prospectively**: training dynamics themselves eliminate redundant neurons.
- **Open**: If we take the clustered network, rewind surviving neurons to initialization, and retrain — does it recover full performance? Are collapsed neurons the “losing tickets”?

Both phenomena point toward a **simplicity bias** in neural network training — the optimizer prefers solutions with lower effective complexity, in both function space and architecture space.

Project 4: Broader Connections — Grokking & Mean-Field

Grokking (Power et al., 2022): Validation accuracy suddenly improves long after training loss saturates.

Bias collapse as a mechanism: The network may first memorize with many neurons (fast, high capacity), then slowly collapse to a sparse solution that generalizes (slow, structural reorganization). The collapse phase corresponds to the grokking period.

Connection to mean-field theory: In the mean-field PDE, bias collapse corresponds to the neuron distribution $\mu(t)$ **concentrating onto a discrete support** — from a diffuse distribution to a sum of Dirac masses. This is a **singularity formation** in the PDE, potentially analyzable with tools from nonlinear PDE theory.